

Enriched Embodiment Environments for Healthcare Spaces: Exploration through the Design of a Cancer Treatment Facility

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Figure 1: A multisensory representation of the patient's breath in the CT simulator

Abstract

The increasing understanding that a patient's experience and mental well-being influence their physical condition, treatment outcomes, and even prognosis, is changing the design of healthcare environments. A gap remains in establishing scientifically grounded methodologies, design guidelines and best practices, for designing real-world healthcare settings that address newly-recognized needs. Drawing on principles from neuroscience and human-computer interaction, we explore the concept of enriched multisensory healthcare environments that employ embodied interaction. This work reviews the principles underlying the design of such spaces, and an implementation of such an environment at a treatment facility within a newly constructed cancer treatment center. Two interaction prototypes were developed, employing embodiment to leverage the benefits of multisensory enriched environments. We detail the design process, decisions, implementation choices, and rationale applied. Finally, we discuss future directions for this work and the enriched embodiment environment approach in healthcare.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; **Mixed / augmented reality**; **Interaction devices**; **User centered design**; • **Social and professional topics** → **Medical technologies**.

Keywords

Healthcare, Enriched environments, Embodiment, Multisensory integration, NeuroHCI, Physiological signals

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1 Introduction

The challenges of designing healthcare environments, interactions, and technologies are of great magnitude and complexity [3, 5, 13, 17, 28, 56, 57]. Currently, the goals of designing healthcare environments are broadening, as the scope of medical treatment is expanding. Today's healthcare spaces must be designed not only to serve logistic and medical procedural needs, but also to aspire to greater goals, such as enhancing the human experience and quality of life [22]. This is becoming increasingly central to modern

healthcare, along with the need for more patient-centered design and interaction [30, 32, 35].

In addition to the human factors attributed to the quality of patient experience, it is now clear that the physical environment can greatly affect mental and physical well-being [6, 17, 18, 34] and that improving the patient's mental state can alter the course of treatment, improving treatment coping and outcome [16, 24, 58]. Elements of the environment contribute to environmental enrichment, which are known to have mental and physical benefits. In this work, we propose the concept of enriched multisensory embodiment environments for healthcare spaces - leveraging the benefits of environmental enrichment by implementing principles of multisensory integration and interactions for inducing embodiment. This in turn maximizes the potential of the environmental enrichment.

We detail a unique use case of an environment designed as a research platform for such interactions. The environment was integrated from the planning stages into the construction of a CT facility, which serves as the first stage of radiotherapy treatment, and is located in a newly constructed cancer treatment wing, designed based on principles of enriched environment architecture [23]. Environmental enrichment is broadly defined as the enhancement of the physical environment with stimulating elements. Enriched environments, providing sensory, cognitive, or social enrichment, have been extensively correlated with numerous mental and physical benefits [2]. Specifically, *multisensory* enriched environments have been shown to be beneficial for mental well-being and emotional health [2, 25, 36, 47, 52]. Enriched multisensory rooms such as the Dutch Snoezelen sensory spaces and others have been shown to benefit populations with cognitive and emotional disorders [7, 8, 11, 25, 40]. Though there are commercial installations that curate audio and visual scenes within hospital settings, such as the Philips Ambient Experience, these employ elements aimed at positive distraction, unlike the current system implemented in this work [48].

Popular contemporary fields of inquiry, such as the theory of embodied cognition and the growing interest in somaesthetics [14, 42, 54] are helping to solidify the idea that there is a bidirectional relationship between our brains, our bodies, and even the environment in which they are situated. This is becoming apparent in neuro/cognitive science [26, 62] but also in HCI and other fields [12, 39, 43, 59]. Embodied cognition proposes that "the human mind is largely determined by the structures of the human body...and its interactions with the physical environment" according to the APA Dictionary of Psychology. In HCI, this concept often relates to the notion of embodiment in virtual environments [4, 38, 45]. VR has been used for mental health benefits [29, 50], yet is limited in hospital uses due to practical limitations. As such, in this work we explore embodiment into features of the extended healthcare environment itself. Dourish states that according to the embodied interaction approach in HCI "the interaction is not only situated, but we are situated in it" [20]. As such, embodiment has been shown in prior research to occur in elements of the extended space, into avatars projected onto screens [60], in teleoperation [46], and in interactive installments [21]. An efficient method to induce a sense of embodiment is physiological interaction, using body signals to profoundly connect phenomenologically to an element or space, for example in real-time interaction such as biofeedback [21, 44, 60].

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Based on the above, we suggest that enriched multisensory embodiment environments could maximize the positive effects of environmental enrichment by inducing embodiment within the healthcare environment itself. Our first implementation of such an environment employed the bodily signal of respiration (see Figure 1), following prior research that used representation of the user's breath (embreathment) to enhance the sense of agency and embodiment in interactive experiences [44, 60]. This biosignal is particularly relevant for interaction in healthcare environments as it can be intentionally controlled, and since breathing techniques and guided breathing have been shown to significantly improve mental and physical states, affect relaxation, arousal, comfort and more [37, 51, 53, 63, 63].

2 Design and implementation of the enriched environment

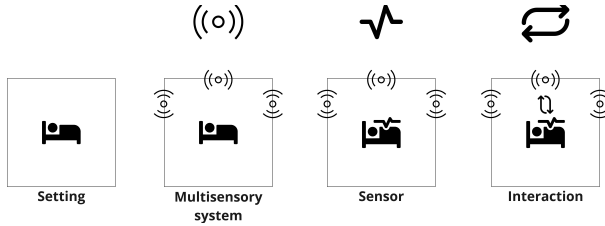


Figure 2: The four elements comprising a multisensory enriched embodiment healthcare environment, supplemented from left to right.

As defined above, an enriched embodiment healthcare environment is one that employs embodied interactions within the setting to maximize the mental and physical enhancements provided through environmental enrichment. Such spaces would be comprised of four main elements - a setting, a multisensory system, a sensor and an interaction (see Figures 2). The *setting* includes all the given medical requirements and space constraints. Incorporating the space and medical equipment, as well as the procedures and any parameters defining the need; The *multisensory system* includes all sensory stimulation elements integrated into the environment (visual, auditory, tactile etc.), and should enable the environment enrichment; The *sensor* would be an element (or multiple ones) capturing data from the patient, to be used in an embodying interaction. In this work, we focus on physiology and body signals, yet this could also involve motion tracking, cameras, or other means; The *interaction* would utilize the different elements to create the embodiment in the environment to achieve the desired mental and physical benefits, including but not limited to: countering negative cognitions, reinforcing positive cognitions, enhancing the sense of control, and other setting specific benefits.

2.1 The setting

This project began with the planned construction of a new cancer treatment wing, that offered a unique opportunity to design an environment in which interactions are integrated as an enrichment factor into the treatment facility itself. The CT facility, referred to as the CT simulator, was chosen because it is the cancer patient's first

encounter with the treatment. The main preparatory procedure that takes place in the facility is a CT scan that visualizes the individual anatomy of the patient. The scan is used for planning the subsequent radiotherapy treatment. This is particularly important, as precise dosing and localization are crucial for achieving an optimal balance between maximal irradiation of the tumor, and maximal avoidance of healthy tissue [1].

We conducted several visits to the site in its construction phases, as well as to a working cancer treatment department in a different hospital, to gain a better understanding of the CT simulation and radiotherapy processes as a whole, and the people involved. We repeatedly interviewed and consulted with different stakeholders, including medical doctors, technicians, construction management and operation teams of the hospital, architects, and the CT manufacturer. Technical discussions also included the sharing of construction plans, medical systems' specifications, and repeated meetings on site with all relevant stakeholders. Due to hospital policies, although we were allowed to view the procedures, we could not interview or pass on questionnaires to patients at this time. Considering the project as analogous to a public space installation, we chose to incorporate a video-art director experienced with public displays into the team from an early stage (Yoav Cohen [15]). He joined on and off site visits and consulted the team as part of an extensive technology review performed to evaluate different solutions against the identified limitations.

As we learned from the medical team, when patients experience this medical setting for the first time, it can induce significant stress. This in turn may affect the patient's physiological functions such as muscle tension, and often leads to differences in positioning between the first scan of the patients and subsequent treatment sessions. This can be detrimental to the success of the treatment, as radiation treatments are planned under the assumption that the exact positioning in the simulation will be replicated in the following radiation therapy treatment sessions [1, 19]. The procedure normally takes place about a week after the doctor's appointment (in which the patient is diagnosed and told they will be undergoing radiotherapy). Overall, the patient spends approximately 20min in the room in which they are positioned, tattooed with three points on their bodies for future localization, and scanned, during which time the patient is alone in the room. We learned that this facility presents a crucial junction for the patients, both with respect to their physical state, and mental, serving as a defining moment in the treatment procedure. The CT room is sometimes the patient's first encounter with such medical devices. Patients are very often stressed about the scan and the coming treatments, and the act of being tattooed often makes them feel very uncomfortable, both physically and mentally.

2.2 The multisensory system

The sensory space designed for this particular setting incorporated elements of the visual, auditory, and tactile senses (see Figures 3) - to enrich the visual experience, a curved LED display was embedded into the ceiling, to create a more immersive experience for the patient within the CT [41, 64]. This was supplemented by atmosphere lights that extend the experience beyond the display to the rest of the room, also allowing a natural transition upon entering

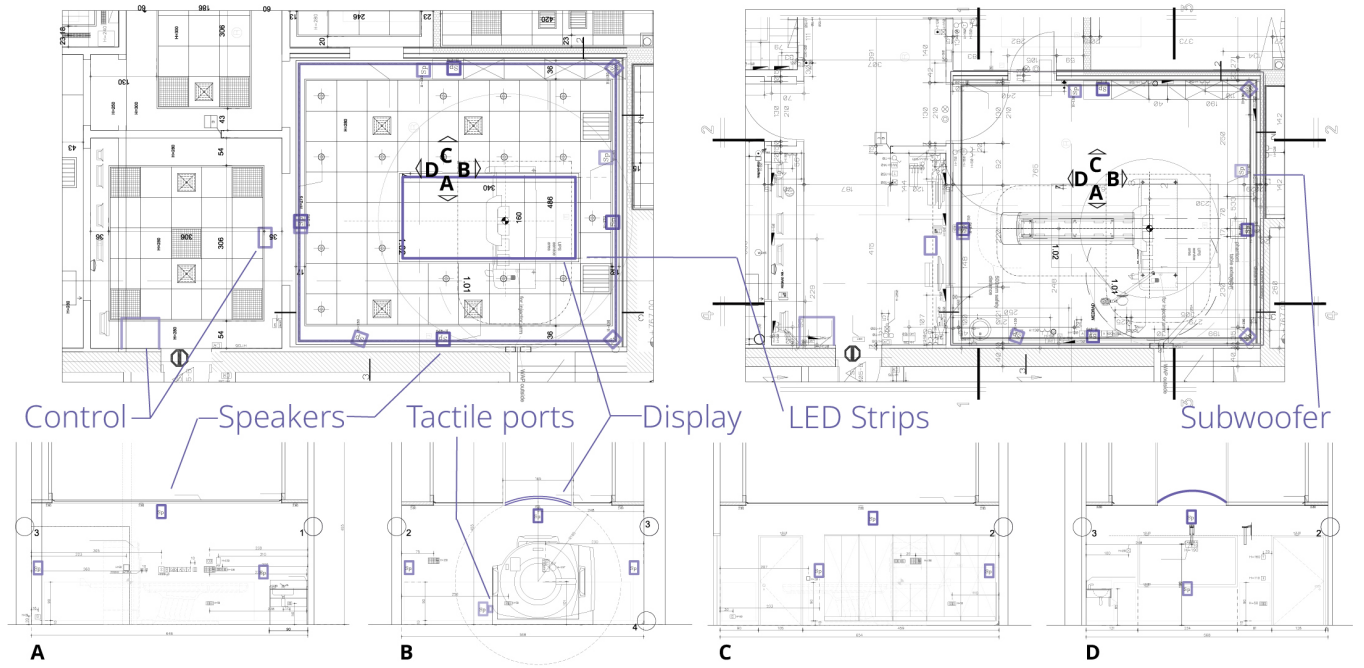


Figure 3: Architectural plans of the system's integration in the facility. The system's elements are colored, opacity is approximately associated with height for clarity in the different views.

the room (see Figures 1). To enrich the auditory experience, we constructed an ambisonic system that allows auditory spaces to be simulated and sounds to be "placed" in space [65, 66]. Lastly, for tactile somatosensory enrichment, we created an infrastructure to allow for the retrofitting of high-fidelity haptic stimulation devices.

The choice of elements was guided by the goal of building a flexible research platform allowing for experimentation with different interactions. This led for example to the choice of elements that are versatile and not bound to a specific use, and the installation of an infrastructure for enhancement in three sensory modalities (though not all were used in our initial prototypes as detailed below). In the design of the system, we aimed to create an immersive environment considering the different stages of the treatment in the space, which would allow for a very high level of control over the patient's sensory experience. Reaching a satisfactory design involved balancing between optimization for immersiveness and control of the environment, while refraining from interfering with factors of the medical procedure. For example, enabling technicians to roam freely around the environment and communicate directly with patients, allowing for handling of treatment accessories, and easy maintenance. Some of the other considerations that affected the design choices included allowing proper ventilation for the CT device, minimizing audio reflections for supporting advanced audio features, equipment location, storage positioning, and more.

Our visual display was a custom-built, 160x336cm curved LED screen with an 860x1806 resolution (see Figures 4). The screen was embedded in the ceiling, and the exact size was determined on the basis of the size of the ceiling panels, taking into consideration other infrastructures embedded in the ceiling. The radius of the display's

curve was determined so as to "surround" the user lying down in the CT (see Figures 3, B). 10 strips of Philips Hue ambient lights, which allow for RGBW gradient lighting were installed in a designated shelter constructed on the ceiling a few centimeters away from the walls, projecting onto them, extending the display's effect to the whole room. To provide auditory stimuli, a unique ambisonic audio system consisting of eight full-range monitor speakers (Genelec 4420A) and one subwoofer (Genelec 7370A) was constructed. An allRad Ambisonics decoder was applied using the spat5 package allowing to apply localized sound in a challenging space for 3D

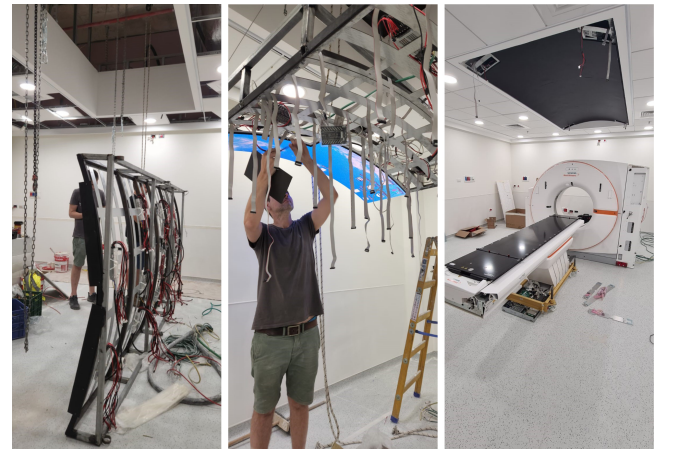


Figure 4: Installation of the curved screen in the CT room.

recreation, due to the combination of uncontrolled reflections, and irregularity of the speaker positioning (due to interfering furniture and devices on the walls, and the large CT device set at the center of the room). Measurements of speaker responses were taken from a number of intended participant head positions and could be defined for best audio reproduction capabilities. With respect to the tactile sense, designated audio output ports were installed next to the head of the CT to allow dedicated haptics to be embedded as part of the multichannel audio system. This setting allows for the delivery of complex, wide bandwidth haptic feedback, naturally correlated with the audio stimuli.

The combined (audio-visual-tactile) multisensory system was designed to be controlled via a dedicated touchscreen control station connected to a computer, audio Dante cards, and visual converters placed in the CT control room. The computer was meant to be used mainly by the research and development team and for processing various experiences, while the touchscreen control station was designed for the medical team to have a simplified interface that allows switching between operation modes (i.e. running a specific experience or completely disabling the system if desired).

2.3 The sensor

To answer the distinctive setting requirements, we developed a wearable, clip on, respiration sensor to fit the specific use case (see Figure 5). As described in the related work section, we found respiration as an especially fitting physiological signal for enhancing embodiment in a healthcare environment. Our choice was also directed by what we learned from the staff about the patients' positioning and later validated in the literature. Namely, that controlled breathing is important for the radiotherapy use case, as stable and slow breathing is desired for enabling stable positioning during treatment, crucial for accurate radiation delivery. Specifically, patients with breast and lung cancer are required to take deep breaths and hold air as part of their treatment, to create a distance between the radiation focus and healthy tissue [55], meaning that uncontrolled respiratory motion can pose a significant risk of damaging healthy tissues during the procedure [61].

The need for designing a dedicated sensor arose after our initial lab tests and first experimentation in the CT simulator, where we used a Biosignals PLUX Piezo-Electric Respiration (PZT) sensor [49]. This sensor, widely used in research, is comprised of an elastic chest belt containing a piezo element, usually placed around the thorax or abdomen, and a Bluetooth hub normally clipped to the patient's clothing. We quickly learned that this sensor has several limitations in our specific context. Beyond some difficulties with recognizing very deep breaths and breathing pauses due to the sensor's technology, the main issues were the hassle and invasiveness

of strapping the belt onto the patient, testing the signal quality, and making adjustments, all of which require handling and can cause discomfort. Another significant issue was that the hospital garment did not afford for attachment of the hub.

Our developed respiration sensor was inspired by prior research [27]. We designed the sensor such that it clips onto the collar of the patient's garment. The sensor utilized an M5StickC PLUS Mini IoT Development Kit, which is a small device that embeds, among other components, an Inertial Measurement Unit (IMU), a battery, and a WiFi component. We built an attachment for the kit to mount over one's collarbone following previous work [33], and developed a code that collects the perpendicular acceleration data from the IMU, filters it, and sends it over WiFi to the operating computer to enable respiration-responsive interactions. Although the sensor has not been evaluated during a CT scan, it should be safe for use to the best of our knowledge, and according to discussions with the facility's chief therapeutic radiographer, its use is not expected to cause artifacts.

2.4 The interaction

The interactions were designed to enhance the environmental enrichment by leveraging embodiment in the enriched environment, and countering negative cognitions, enhancing agency and body awareness. Specifically for this setting, controlling and lowering respiration rate was also desired. We present here two first prototypes exploring our proposed approach, and as initial validation of the environment as a platform.

2.4.1 Interaction 1. The first prototype using the system was centered around a 3D virtual bubble avatar that simulates breath by descending and ascending accompanied by ambient music (see Figure 6). The prototype development started with experimentation in the lab. A series of interactions were experimented with, involving a representation of the user's breath manipulated to guide or encourage desired breathing patterns, surrounding themes explored in the previous workshoun to enhance flow, interoceptive sensibility and body awareness [31]. The initial tests, later adapted to the facility, used an abstract avatar that represents one's breath (see Figure 1). The latest lab iteration which inspired this design, included an avatar expanding on inhalation and contracting on exhalation based on the user's breath, accompanied by ambient music that also reacted to the breath. Following these initial experiments, a brief was formed, and a variation of the concept was developed for the CT simulator in collaboration with the video-artist, in which the bubble shaped avatar was designed to descend on inhalation, and ascend on exhalation. Ascending was accentuated, creating in practice an ongoing movement upwards in the open space, controlled by the breath. This adjustment to the original concept was made with the desire of embodying the user in an open, peaceful space, and allowing them a sense of movement and control while they are required to limit their movement during the CT procedure. The vertical movement also lent itself better to the vertical screen in the space. Some visual elements in the surroundings, such as the sky, the sun and a passing hot-air-balloon also changed over time to enhance engagement.



Figure 5: The custom-made respiration sensor.

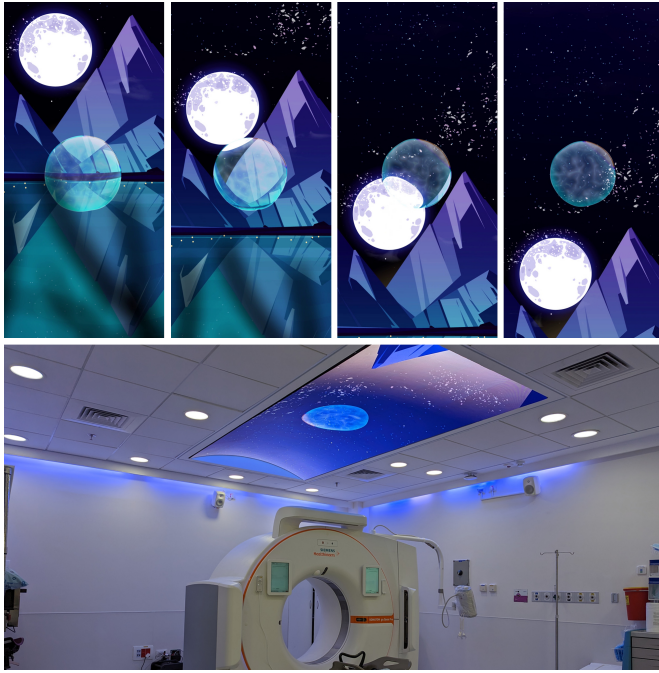


Figure 6: Interaction prototype I - The bubble is designed to descend on inhalation, and ascend on exhalation.

This first prototyping process yielded meaningful insights to guide the design of future implementations. Namely, from our experimentation in the CT we better understood the practical limitations on the available viewing field within the CT simulator device, which vary throughout different stages of the procedure. We also found that integration with the hospital's existing breath sensors would require a more complex solution involving its manufacturer, prompting us to opt for the use of standard kit sensors [49]. This is also where we learned about the limitations of using a belt-based sensor in the treatment environment, which ultimately drove us to develop our own in-house solution as described above.

2.4.2 Interaction II. Our second interaction prototype iterated on the first interaction following the insights gained. As such, the prototype was adapted to the limitations of the screen's viewing angles in some stages of the procedure by designing an avatar that moves on the different parts of the screen and also leverage the auditory element to give it a presence when not visible. The avatar design consisted of a group of glowing particles, moving throughout the ceiling space of the room, frequently crossing the screen, leaving a trail of glowing particles behind it (see Figure 7). The glow of the particles corresponded to the user's inhalation and exhalation, with the brightness of these particles changing as the patient breathed in and out. The spread of the particles on the screen, together with the ambient light in the room and the ambisonic localized sound system that corresponded to the position of the avatar, allowed a continuous experience of the interaction, clearly noticeable from every position. The background of the visual scene displayed a video of tree-tops moving in the wind, appropriate for the user's viewing



Figure 7: Interaction prototype II - The particles move around the space, crossing the screen while leaving a fading trail. The glow increases on inhalation and decreases on exhalation.

angle, and was based on the proven effect of nature on patients' health [9, 10, 58]. An audio track of matching natural sounds was created, with different sounds emanating from different locations in space by employing the ambisonics system for spatial audio. From a technical standpoint, the data received from the patient's breathing belt underwent signal processing in Python and MAX/MSP. This processing served to eliminate signal noise and achieve real-time smoothing, enabling a seamless animation. The refined signal was transmitted to Unity, where it was translated into the particles' glow level displayed on the screen. Simultaneously, the position of the particle system was transmitted back to MAX/MSP, facilitating the calculation of sound source positions in alignment with the patient's perspective, utilizing the unique capacities of our system.

3 Discussion

3.0.1 Developing the current implementation. Going forward, the system should be assessed in the day to day functions of the health-care environment. Some of the aspects to be evaluated include not only the psychological and cognitive effects specifically mentioned above, but also parameters such as hope, sense of being seen and heard, as well as physiological parameters such as hormone levels, pain and discomfort, and in the long run - treatment outcomes. In parallel, the effects of the system on staff should be evaluated, such as their satisfaction and engagement, while refining the practical operational aspects. The insights gained from such a process could greatly contribute to further advancement of the principles and design guidelines for the development of other future environments.

3.0.2 Expanding the scope of implementations. Beyond the implementation presented, enriched embodiment environments can be developed to offer various implementations and solutions in healthcare settings. Some other healthcare spaces that could lend themselves well to such interventions include radiotherapy facilities, MRI devices, recovery rooms, and delivery rooms. The approach could also be extended to waiting rooms, staff resting areas, or healthcare spaces not situated in hospitals, and elective or alternative healthcare environments. While a wide range of sensors, multisensory systems and interactions can be explored for these various settings, a few overarching future concepts were identified: Connecting between different settings (e.g. creating a similar implementation as the one created for the CT simulator in the radiotherapy facility, linking between the two environments); Creating continuity between recurring visits or stages in treatment over time (e.g. creating a progressing interaction that develops between the recurring visits to the radiotherapy facility); Mediating and mitigating the medical procedure taking place by employing body awareness (e.g. real-time representing the radiation and its effect on the cancer cells in the embodied interaction); and adding elements of personalization (e.g. personal adjustment of an avatar or other elements in the environment).

4 Conclusion

In this work we presented the concept of enriched embodiment environments for healthcare spaces, as environments that employ embodiment for leveraging the benefits of multisensory enriched environments. We provided the rationale and grounding for this approach, defined the basic components for its application, and explored the concept through the implementation of such an environment in an operational healthcare facility. Through this work, we set the foundation for the development of methodologies, design guidelines and best practices, for designing multisensory enriched embodiment environments in healthcare spaces.

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